

tf.cond Stay organized with collections

Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/cond

Return true_fn() if the predicate pred is true else false_fn().

Function Signatures

```
true_fn()
```

```
false_fn()
```

```
tf.cond( pred, true_fn=None, false_fn=None, name=None )
```

Code Examples

```
tf.cond(  
    pred, true_fn=None, false_fn=None, name=None  
)
```

```
tf.cond(  
    pred, true_fn=None, false_fn=None, name=None  
)
```

```
@tf.function  
def fun1(x,y):  
    if x > 0: # AutoGraph converts if-statement to tf.cond().  
        z = y+1  
    else:  
        z = y-1  
    return z  
fun1(tf.constant(7), tf.constant(3)).numpy()  
4
```

```
@tf.function
```

```
def fun1(x,y):
```

```
if x > 0: # AutoGraph converts if-statement to tf.cond().
```

Documentation

Return true_fn() if the predicate pred is true else false_fn().

Used in the notebooks

- Extension types
 - Tutorial on Multi Armed Bandits in TF-Agents
 - Sending Different Data To Particular Clients With `tff.federated_select`
 - End to end example for BigQuery TensorFlow reader
 - Instance Segmentation with Model Garden

For more information, see `tf.function` and AutoGraph guide.

`true_fn` and `false_fn` both return lists of output tensors. `true_fn` and `false_fn` must have the same non-zero number and type of outputs.

Although this behavior is consistent with the dataflow model of TensorFlow, it has frequently surprised users who expected a lazier semantics.

Consider the following simple program:

If $x < y$, the `tf.add` operation will be executed and `tf.square` operation will not be executed. Since `z` is needed for at least one branch of the `cond`, the `tf.multiply` operation is always executed, unconditionally.

Note that `cond` calls `true_fn` and `false_fn` exactly once (inside the call to `cond`, and not at all during `Session.run()`). `cond` stitches together the graph fragments created during the `true_fn` and `false_fn` calls with some additional graph nodes to ensure that the right branch gets executed depending on the value of `pred`.

`tf.cond` supports nested structures as implemented in `tensorflow.python.util.nest`. Both `true_fn` and `false_fn` must return the same (possibly nested) value structure of lists, tuples, and/or named tuples. Singleton lists and tuples form the only exceptions to this: when returned by

true_fn and/or false_fn, they are implicitly unpacked to single values.

Returns

Raises

Example: